Trustworthy and Explainable Agents for Phishing Detection with Grounded Cross-Model Evidence Agreement

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Abstract

When a phishing detector marks a website, the verdict drives a browser warning, a blocklist entry, or a take-down request, so the operator who acts on it needs a reason they can check and a rule for when the verdict is safe to trust on its own. Large language models now reach high accuracy here, yet they emit a *label* and discard the *reason*, and the usual reliability signals – model confidence, self-consistency, and majority voting – read trust at the label, where several models can agree on the right verdict while relying on different or even fabricated evidence. To close this gap, we propose `PhishProof`, a selective phishing detector that reads trust from grounded cross-model evidence agreement instead of from labels. A small panel of diverse language-model agents investigates each page, plans which of its URL, DOM, certificate, redirect, screenshot, and logo signals to gather, and calls verification tools – including existing reference-based and LLM detectors – to confirm every claim against the page itself. `PhishProof` acts only when the agents agree on cues the page confirms and otherwise abstains, returning those agreed, verified cues as a checkable explanation when it does act; a held-out labeled split calibrates the agreement-and-groundedness score to a probability of correctness and sets the operating point. Because the panel reads the page like the specialized detectors do, it stays competitive with them on full-coverage accuracy while adding the calibrated abstention and verifiable explanation they lack. On a phishing benchmark of feed-sourced phishing pages paired with legitimate login pages, `PhishProof` attains, by a clear and statistically significant margin, the best-calibrated trust score and the highest selective accuracy of any method, so an operator can read its reported probability of correctness directly; its area-under-risk–coverage is the lowest in point estimate but lies within the confidence interval of the strongest baseline, and it ranks risky verdicts more reliably than the label-agreement baselines, and on the harder lookalike pages it abstains more often on the fragile right-verdict-but-wrong-reason cases, though that effect is modest where the off-brand signal is unambiguous.

Keywords: phishing detection, LLM agents, tool-augmented agents, evidence cues, selective prediction, cross-model evidence agreement, explanation faithfulness, abstention, trustworthy ML

1. Introduction

A phishing verdict is a consequential act: marking a page as phishing drives a blocklist entry, a browser warning, or a take-down request that pulls a live service off the web. Large language models have sharply raised the accuracy of such detectors, reading a page's URL, DOM, rendered screenshot, and certificate to decide whether it impersonates a brand it does not own [1, 2]. Yet accuracy is only half of what a deployment needs: the operator who must justify a take-down and the user who sees a warning both need a *reason* they can check, and the system needs a criterion for *when* a verdict is trustworthy enough to act on without review. For a decision that can be contested, the bottleneck is no longer whether the model is usually right, but whether a particular verdict can be *trusted and explained*.

Existing detectors approach this only indirectly. One line runs a single LLM over the page and reports a class probability or a verbalized confidence [1]; reliability is then a property of that one model's calibration [3]. A second line aggregates several models or several samples – self-consistency over one model [4], majority voting over an ensemble [5], or weighted label agreement in the style of crowd-sourcing and label-free evaluation [6, 7] – and reads reliability off how often the votes concur. A third, recent line builds multi-agent detectors whose agents specialize on different signals and whose rationales are merged into a single explanation [8, 9]. Together these families cover much of the space of accuracy and aggregation.

They share a blind spot, visible in a routine case (Fig. 1). A page impersonating a bank is shown to a panel of three diverse models; all three return "phishing," so every aggregation method above reports high confidence. But the models reached that verdict for different reasons: one cited the morphed logo, one cited a credential form posting off-domain, and one cited a certificate mismatch that does not actually hold on the page. First, the detectors emit a label and discard these reasons, so

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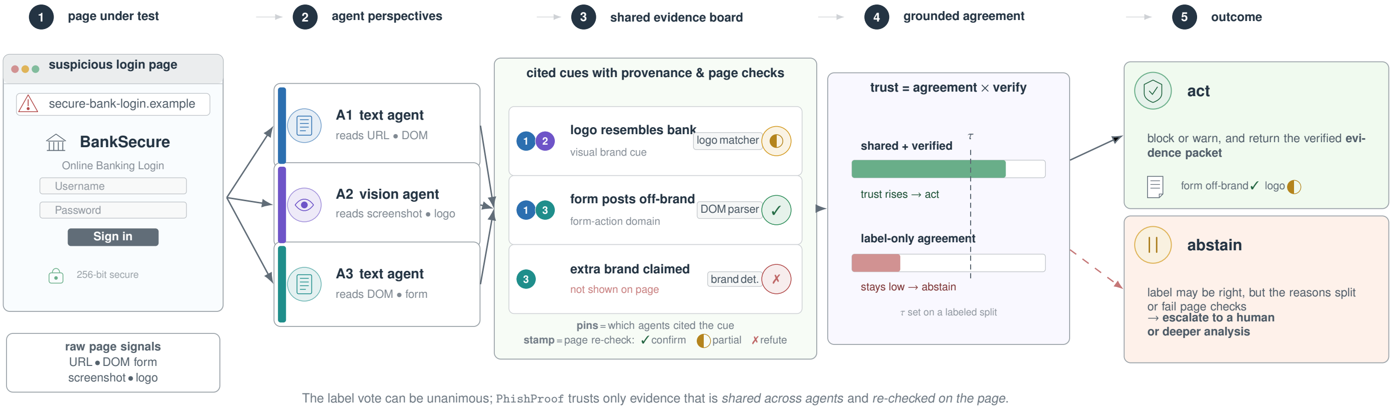

Figure 1: `PhishProof` as a qualitative evidence story. Three agents inspect the same page from complementary views, then place their cited cues on a shared evidence board. Colored pins show provenance, while tool badges mark cues that are confirmed or rejected by the page itself. `PhishProof` acts when the shared reasons survive these checks and returns them as an evidence packet; when agents agree on the label but their reasons split or fail verification, it abstains.

nothing auditable reaches the operator (**L1**). Second, reliability is read at the *label*, where the three agree, and is therefore blind to the fact that they *disagree on the evidence*: the verdict is right for the wrong reason (**L2**). Third, even where the models do agree on a reason, that agreement can be a shared hallucination of a cue that is not on the page (**L3**). A right-label, wrong-reason verdict is not a curiosity: it is precisely the verdict a small change to the cited cue would flip, and precisely the one an operator should not act on blindly.

Two properties of phishing, not shared by generic question answering, make a direct remedy possible. First, the evidence a detector relies on is *objectively checkable*: a form-action domain, a certificate organization, a redirect target, and a rendered logo can each be re-derived from the page itself. Second, labeled pages are cheap to assemble – public feeds publish freshly confirmed phishing URLs and legitimate sites supply benign pages – so a detector's operating point can be set on a held-out labeled split rather than guessed. The evidence that decides a verdict is also *heterogeneous*, spanning several model types (text LLMs and a vision-language model), many signals (URL, DOM, certificate, redirect chain, logo, and credential intent), two modalities, and a verification mechanism per signal. Deciding which signals to gather for a given page and how to combine them is a *planning-and-aggregation* problem better suited to an agent than to a fixed pipeline, and modern LLMs can act as such agents – emitting typed outputs and calling tools to gather and verify their own evidence. The opportunity is to stop treating reliability as a property of labels and start treating it as a property of *evidence that several independent agents plan, gather, verify, and agree on, and that the page confirms*.

To this end, we present `PhishProof`, a multi-*agent* phishing detector that trusts a verdict in proportion to grounded cross-model agreement on the evidence. A small, deliberately diverse panel of LLM *evidence agents* investigates the page: rather than emit a bare label, each agent *plans* which signals and modalities to examine, then gathers and grounds its evidence by calling verification *tools* – a DOM parser, a certificate reader, a redirect tracer, a logo matcher, and an existing reference-based or LLM brand detector – and records what it found in a shared, per-page set of typed *evidence cues*, each carrying the asserting agent and the confirming tool. `PhishProof` *aggregates* these grounded findings: it measures agreement at the level of the cited *evidence* rather than the label, weights that agreement by whether each agreed cue verifies against the page, and combines the two into a single Grounded Evidence-Agreement score. It acts only when the score is high – returning the agreed, verified cues as a checkable explanation – and otherwise abstains, escalating the page to deeper analysis or a human. A calibrator fit on a held-out labeled split maps the score to a probability of correctness, so the operating point is set against real outcomes rather than guessed. Reusing the existing detectors as tools also makes `PhishProof` competitive on full-coverage accuracy by construction, since the panel inherits their signal. The novelty over prior multi-agent detectors is the unit of agreement: not the label, and not one model's introspective faithfulness, but a set of evidence cues that is simultaneously cross-agent-agreed and tool-verified.

Realizing this raises three difficulties that the design must rule out. First, to answer **L2**, agreement over structured evidence must be defined so that it cannot collapse into label agreement in disguise – two models can name the same brand while resting on entirely different cues. Second, to answer **L3**, shared hallucination must be caught cheaply, by checking cited cues against the page rather than trusting the models to police themselves. Cutting across both, the panel must stay small: every added agent is another round of model and tool calls, so the gain has to come from agent *diversity* rather than count.

We make the following contributions.

- **Failure mode.** We identify the *right-label, wrong-reason* failure of multi-model phishing detectors and cast trustworthy detection as selective prediction over an evidence-agreement signal (§3).
- **Method.** We introduce `PhishProof`, a panel of tool-using evidence agents that cite a shared set of typed, verified evidence cues, and its Grounded Evidence-Agreement score, which measures cross-agent agreement over those cues and turns it, calibrated on a held-out labeled split, into a

selective act/abstain decision with the agreed, verified cues as the explanation (§4).

- **Evidence.** On a phishing benchmark of feed-sourced phishing pages paired with legitimate login pages, `PhishProof` is competitive on full-coverage detection while improving the risk–coverage trade-off over single-model confidence and label voting: it has, by a clear and statistically significant margin, the lowest expected calibration error of any method (0.015) and the highest selective accuracy at 80% coverage (97.0%), so its reported probability of correctness can be acted on directly; its AURC is the lowest in point estimate (38% below the label-agreement baseline) though within the confidence interval of the strongest baseline (§5).
- **Analysis.** We isolate the contribution of evidence-level agreement, tool-grounding, agent diversity, and calibration; cross-family *diversity* and calibration are the dominant factors, while on this benchmark – where phishing pages carry an unambiguous off-brand signal and few shared hallucinations – grounding adds little to the ranking and the right-label-wrong-reason effect is modest, appearing only on the harder lookalike pages (§5, §6).

The rest of the paper is organized as follows. §2 positions `PhishProof` against label-level reliability and multi-agent detection. §3 formalizes grounded evidence-agreement and the selective objective. §4 details the evidence agents, the cues they cite, the score, and the calibrated decision rule, and §5 reports the evaluation.

2. Related Work

LLM and reference-based phishing detection. The dominant modern detectors decide whether a page impersonates a brand it does not own. Reference-based systems match a rendered page against a brand library of logos and layouts [10, 11, 12], and recent work refreshes that reference set online [13] or replaces it with an LLM that recognizes the brand and checks the domain directly [1, 2]. A parallel line treats the page as URL or image features for a learned classifier [14, 15], surveyed broadly [16]. Across these sub-mechanisms the shared target is *aggregate* accuracy, reported as a single page-level verdict. Building on this line rather than competing with it, `PhishProof` invokes such detectors as verification tools (§4.3) – inheriting their accuracy so that it stays *competitive* at full coverage (Tab. 4) – and adds what a bare detector lacks: a *per-verdict* decision about whether the verdict can be trusted, acted on automatically, and explained (§4.4).

Multi-agent and ensemble phishing detection. Closest to us are detectors that run several models. Majority voting over LLMs has been studied directly for phishing [5]; multi-agent systems assign agents to different signals and either debate to a verdict or merge their rationales into one explanation [8, 9], sometimes backed by an episodic memory of past cases [17]. Unlike these systems, which aggregate or consolidate at the level of the *label* or a fused rationale, `PhishProof` measures agreement at the level of individually cited evidence units ((2)) and treats *disagreement on evidence* as the signal to abstain rather than a conflict to resolve. Its agents are tool-using investigators that each *plan* their own investigation across signals and modalities, and whose tool-grounded findings are *aggregated* (§4.3) into a shared set of cited evidence cues that serves as both the trust signal and the explanation, rather than a fused free-text rationale.

Label-level reliability and label-free evaluation. A verdict's reliability is usually read from the label distribution: post-hoc confidence calibration [3], agreement across samples of one model [4], or agreement across many models, as in crowd-sourcing aggregation [6], agreement-on-the-line under shift [18], and trust calibration for phishing detectors [7]. Where these estimate reliability from *how often labels concur*, `PhishProof` estimates it from *whether the reasons concur and hold*, which separates pages on which the panel agrees by coincidence from those on which it agrees because the page truly carries the evidence ((4)).

Explanation faithfulness. Causal audits show that an LLM's stated rationale is often not the driver of its decision [19], and multi-agent debate has been used to *improve* the faithfulness of generated explanations [20]. Such audits operate on a single model and require counterfactual re-execution. In contrast, `PhishProof` uses cross-model agreement on grounded evidence as an inexpensive faithfulness *proxy*: a reason that several independent models cite and that the page confirms ((3)) is, by construction, less likely to be a post-hoc rationalization, at no extra re-execution cost.

Selective prediction. We build on selective classification, which trades coverage for accuracy via an abstention rule [21], and on distribution-free uncertainty quantification [22]. Building on this foundation, `PhishProof` supplies the missing ingredient for the phishing setting: an abstention score derived from grounded cross-model agreement ((4)), calibrated on a held-out labeled split so the risk–coverage operating point is set against real outcomes (§4.4).

Position and scope. `PhishProof` contributes a single new object, the Grounded Evidence-Agreement score, and the selective detector built on it. Tab. 1 positions it against one representative per family along five capabilities – evidence-level agreement, page-grounding, calibrated trust, selective abstention, and a verifiable explanation as output – and `PhishProof` is the only entry that supports all five. We focus on per-verdict trust and explanation for a multi-model phishing detector; the framework is agnostic to the specific base models and evidence schema. We do *not* claim (i) a new base detector or a leaderboard win on full-coverage accuracy – `PhishProof` reuses existing detectors as tools and is *competitive* with them (Tab. 4), contributing the per-verdict trust and explanation layer on top; (ii) robustness to an adversary who controls the panel agents or corrupts the tools' inputs, a harness-level threat we treat as out of scope; or (iii) coverage guarantees on fully cloaked pages that carry no verifiable cues, where the detector necessarily abstains rather than errs.

Table 1: Capability comparison for per-verdict trust in multi-model phishing detection. ●: yes; ◑: partial; ○: no. Columns are defined in the body; full-coverage detection accuracy is reported separately in Tab. 4.

Approach	Evid.-level	Page-grounded	Calib.-trust	Selective abstain	Explan. output
Phishpedia [11]	○	◑	○	○	◑
PhishLLM [1]	○	◑	○	○	◑
Single conf. [3]	○	○	◑	●	○
Self-consist. [4]	○	○	○	◑	○
Majority vote [5]	○	○	○	◑	○
Label agree. [6]	○	○	◑	◑	○
MultiPhishGuard [9]	◑	○	○	◑	◑
Faithfulness audit [19]	◑	○	○	○	◑
`PhishProof` (ours)	●	●	●	●	●

Table 2: Summary of important notations.

Symbol	Meaning
x	a webpage under test (URL, DOM, screenshot, cert, redirects)
$\mathcal{M} = \{m_1, \dots, m_M\}$	panel of M diverse LLM evidence agents (M=3)
$\mathcal{U}$	fixed typed evidence-cue schema (§4.1)
$E_i \subseteq \mathcal{U}$	grounded cues agent m_i cites on x
C	*consensus* cue set (cues a majority of agents assert)
$g(u, x) \in [0, 1]$	tool that re-derives cue u's type from x
$A(x) \in [0, 1]$	evidence-level agreement over the shared cue set
$G(x) \in [0, 1]$	groundedness of the consensus cues
$\mathrm{GEA}(x) = A \cdot G$	Grounded Evidence-Agreement score
τ	abstention threshold (acts iff GEA $\geq \tau$)
$\hat{p} = \kappa(\mathrm{GEA})$	calibrated map GEA $\to \mathbb{P}$(correct), fit on a labeled split

3. Problem Formulation

We formalize trustworthy phishing detection as *selective prediction over a grounded evidence-agreement score*: rather than always emitting a label, the detector acts only when its evidence is both agreed-upon and verified, and otherwise abstains. We proceed in five steps – what the detector observes (§3.1), the score it computes (§3.2), the act/abstain objective it optimizes (§3.3), the requirements the method must meet (§3.4), and the resulting framework (§3.5). Tab. 2 collects the notation.

3.1. Preliminaries

Setting. A detector receives a webpage x – represented by its URL, DOM, rendered screenshot, TLS certificate, and redirect chain – and must decide whether x is phishing. It has access to two resources. The first is a panel $\mathcal{M} = \{m_1, \dots, m_M\}$ of M *diverse* LLM *evidence agents*: they come from different model families and include at least one vision-language model that reads the screenshot, and each can call a set of verification *tools* that re-derive factual signals from x. The second is a fixed, typed *evidence schema* $\mathcal{U}$ that fixes the vocabulary of facts the agents may cite. A *cue* $u = (t, v) \in \mathcal{U}$ pairs a type t (e.g. form-action-domain, cert-org, redirect-target, logo-brand, credential-intent) with a value v (e.g. the domain a login form posts to); every type has a corresponding tool g that re-derives it from x, which is what makes a cue checkable rather than merely asserted.

Agent outputs and the shared cue set. Run on x, each agent investigates the page and returns a verdict together with the cues it grounds through tool calls:

$$m_i(x) = (y_i(x),\ E_i(x)), \quad y_i \in \{\mathsf{phish}, \mathsf{benign}\},\ E_i \subseteq \mathcal{U}. \quad (1)$$

Pooling the agents' cited cues gives the page's *shared cue set* $\bigcup_i E_i(x)$, the universe of evidence anyone raised, and the verdict is the majority vote $\hat{y}(x) = \mathrm{maj}_i\, y_i(x)$. Within the shared set, a *consensus* cue is one that a strict majority of the agents assert with a consistent value, $C(x) = \{u : |\{i : u \in E_i\}| > M/2\}$ – the evidence the panel independently corroborates; these consensus cues are the evidence `PhishProof` trusts to drive the verdict and returns as the explanation.

3.2. Grounded Evidence-Agreement

Existing detectors estimate per-verdict reliability from the *label* channel alone – a single model's confidence, or the concordance of the verdicts $\{y_i\}$ [3, 5, 6]. The flaw is that the label channel is blind to *why* the agents agree. We instead score the *evidence* channel with two complementary factors: how much of the cited evidence the panel agrees on, and how much of that agreed evidence the page actually bears out. Formally, the evidence-level agreement and the groundedness of the consensus are

$$A(x) = \frac{|C(x)|}{\left|\bigcup_i E_i(x)\right|}, \quad (2)$$

$$G(x) = \frac{1}{|C(x)|} \sum_{u \in C(x)} g(u, x), \quad (3)$$

where $g(u, x) \in [0, 1]$ is the tool that re-derives cue u's type from x (deterministic, returning 0 or 1, for form-action, cert-org, and redirect; perceptual, returning a similarity, for logo-brand), and we set $G(x) = 0$ when $C(x)$ is empty. Thus A rises only as the agents converge on the *same* cues, and G rises only as those agreed cues survive re-derivation from the page. The *Grounded Evidence-Agreement* score is their product,

$$\mathrm{GEA}(x) = A(x) \cdot G(x) \in [0, 1]. \quad (4)$$

The product, rather than a sum or average, is deliberate: either failure alone drives GEA to zero. Agreement on cues the page does not confirm sends $G \to 0$, and a panel that concurs on the label while citing disjoint cues sends $A \to 0$; only when the agents agree on the evidence *and* the page bears it out does GEA stay high. This is exactly the conjunction the label channel cannot express.

3.3. Selective Objective

`PhishProof` is a *selective* detector: it acts with the verdict $\hat{y}(x)$ when $\mathrm{GEA}(x) \geq \tau$ for a threshold τ, and otherwise abstains, emitting $\perp$ to escalate the page. A threshold induces two quantities – the *coverage* $\phi(\tau)$, the fraction of pages on which the detector acts, and the *selective risk* $r(\tau)$, the error rate on exactly those pages:

$$\phi(\tau) = \mathbb{P}[\mathrm{GEA} \geq \tau], \quad r(\tau) = \mathbb{P}[\hat{y} \neq y \,|\, \mathrm{GEA} \geq \tau], \quad (5)$$

with y the true label. Sweeping τ traces the risk–coverage curve $r(\phi)$, and a good score keeps this curve low everywhere – equivalently, it minimizes the area under it (AURC). To turn the raw score into an actionable threshold, we fit a calibrator $\kappa : \text{GEA} \mapsto \hat{\mathbb{P}}(\hat{y} = y)$ on a held-out labeled split that maps the score to a probability of being correct, and pick the operating point τ that realizes a target risk. The per-page *output* is then a verdict or an abstention, the calibrated trust $\hat{\rho} = \kappa(\text{GEA})$, and, when acting, the consensus cue set $C(x)$ as the explanation.

3.4. Design Requirements

Scoring the evidence channel rather than the label channel imposes three requirements, each repairing one limitation of label-centric reliability identified in §1. The framework (§3.5) discharges each through a named component, and §5 measures its effect.

(**R1**) *Emit structured, checkable evidence.* The label channel returns no auditable artifact, only the verdict $\hat{y}$, so an action cannot be justified or verified (answers **L1**; realized by agentic evidence elicitation, component C_1, §4.1, and returned as the explanation throughout).

(**R2**) *Measure agreement at the evidence level.* Concordance read from the verdicts $\{y_i\}$ cannot distinguish a panel that agrees *because* the page carries the evidence from one that agrees on the label while resting on entirely different cues (answers **L2**; realized by evidence agreement A, component C_2, §4.2; ablated in §5.4).

(**R3**) *Ground cited evidence to the page.* Evidence the panel cites in agreement may still be a *shared* hallucination, which agreement alone would reward; it must instead be checked against x (answers **L3**; realized by tool-grounding G, component C_3, §4.3; ablated in §5.4).

This limitation–requirement–component thread runs through the paper: each C_i is developed in §4 and isolated by an ablation row in §5.4, so the contribution of every requirement is measured, not assumed.

3.5. Framework Overview

End-to-end, `PhishProof` turns a page into a trusted-and-explained verdict in four steps, drawn in Fig. 2 and developed one per subsection in §4; the four components C_1–C_4 map one-to-one onto these steps. **(1) Investigate.** Each agent plans which signals to examine and calls the verification tools, emitting its verdict and a set of typed, tool-checkable cues into the shared cue set (C_1 agentic evidence elicitation, **R1**, §4.1). **(2) Aggregate on evidence.** Over the shared set we form the strict-majority consensus and measure how much of the cited evidence falls within it – the evidence-level agreement A – instead of counting how often the verdicts concur (C_2, **R2**, §4.2). **(3) Verify against the page.** Each consensus cue is re-derived by its tool, and the groundedness G discounts any consensus that the page does not actually bear out, catching shared hallucinations (C_3, **R3**, §4.3). **(4) Decide and explain.** The score $\text{GEA} = A \cdot G$ is calibrated on a labeled split and thresholded into an act/abstain decision; when it acts it returns the consensus cues as the explanation, and otherwise escalates the page (C_4 calibrated selection, §4.4). Because the score is a product, a verdict survives to action only when steps (2) and (3) both hold – precisely guarding against the right-label/wrong-reason and shared-hallucination failures the label channel misses.

4. The `PhishProof` Detector

`PhishProof` turns the Grounded Evidence-Agreement score of (4) into a calibrated, selective verdict through the four components C_1–C_4 previewed in §3.5 and drawn in Fig. 2. The four subsections below develop them in the order a page flows through the system – elicit evidence, agree on it, ground it, then decide – after which we give the end-to-end procedure, its guarantee, and its cost.

4.1. Evidence agents and the shared cue set

This component produces the per-agent outputs of (1). Each member of the panel is an *evidence agent*: prompted with the page, it *plans* which signals and modalities to examine, calls the verification tools to obtain them, and returns a verdict together with the grounded cues it found, drawn from the fixed schema $\mathcal{U}$ and emitted by constrained decoding as well-typed (t, v) pairs rather than free text (**R1**). Constraining the output to the schema is what makes the cue auditable: it is a typed claim a tool can later re-derive, not prose to be re-interpreted. The schema is small and auditable – the brand claim, the form-action domain, credential intent, the logo brand, a brand–domain consistency cue derived from the agreed brand and the host, and, where the capture provides them, the certificate organization and redirect target (both absent in the corpus of §5, hence inert there) – and is shared by every agent, so their cues are directly comparable. The cues are pooled into the page's *shared cue set*, where each cue is typed, tool-checkable, and tagged with the agent that asserted it, so the verdict rests on a list of cited, verifiable reasons – a brand claim paired with off-brand infrastructure – rather than an opaque score. Agent *diversity* is a design choice, not an afterthought: the three agents come from different model families and at least one is a vision-language model that reads the screenshot, so that *agreement* on a cue reflects independent corroboration rather than a shared prior the agents happen to hold (its value is isolated in §5).

4.2. Evidence agreement

This component scores the shared cue set of §4.1. To *aggregate* and compare cues across agents, two agents must be able to assert the "same" cue despite surface differences, so we first *normalize values*: domains are reduced to their registrable eTLD+1 and brands to a canonical lexicon entry, so that "paypal-secure.com/login" and "paypal-secure.com/signin" count as one form-action cue. On the normalized cues, the consensus set $C(x)$ collects those a strict majority of the agents assert, and the evidence-level agreement $A(x)$ of (2) is the share of all cues raised that fall in this consensus (**R2**). One refinement keeps the rule fair to single-source evidence: a *perceptual* cue visible to only one modality – the logo, which only the vision agent can read – is admitted to the consensus on its own, since

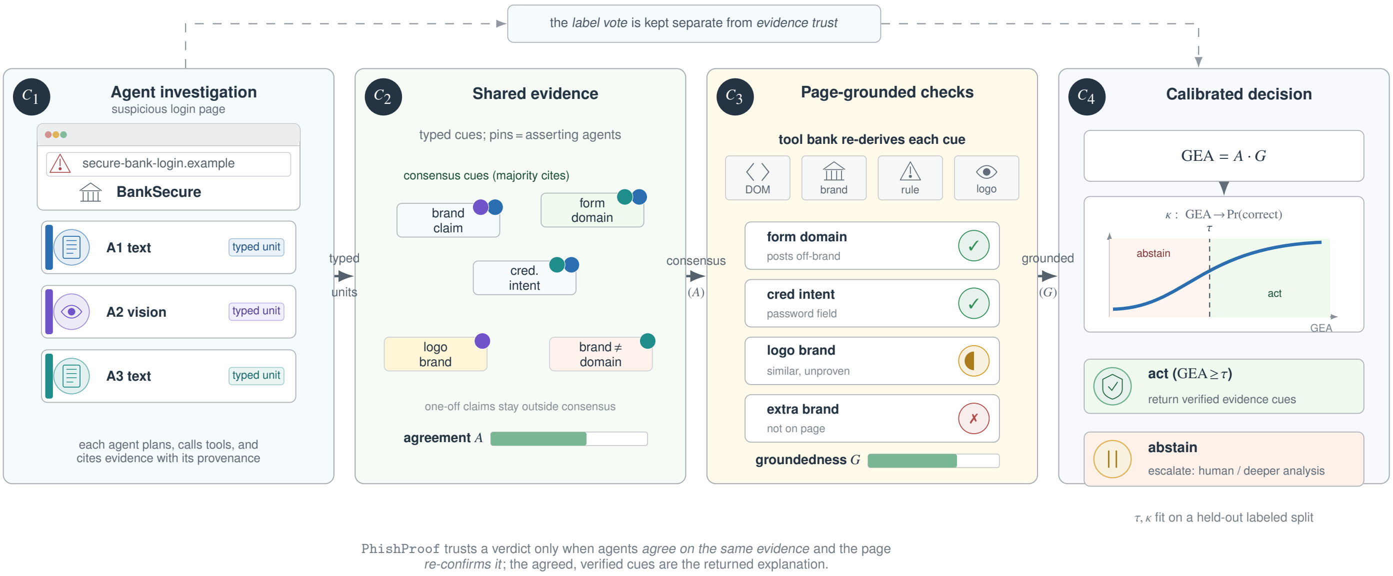

Figure 2: `PhishProof` qualitative pipeline. Agents inspect the same page from complementary views and cite typed evidence cues into a shared set. The system trusts a verdict only when the agents converge on the same cues and the tool bank can re-check those cues against the live page. The calibrated selector returns the label with the verified cues when the evidence is shared and grounded; otherwise it abstains, including cases where the label vote is plausible but the reasons are thin or unverified. The threshold and calibrator are set on a held-out labeled split.

it cannot reach a majority when a lone agent sees the screenshot; every other type still requires a strict majority. Agreement is therefore high only when the agents converge on the *same* evidence, and – unlike a label-vote concordance – is unmoved by their merely concurring on the verdict.

4.3. Tool-grounding

This component verifies the consensus cues from §4.2 against the page. Agreement alone cannot catch a *shared* hallucination – a cue the whole panel asserts but the page does not carry – so each consensus cue is re-derived from the page by the tool for its type, $g(u, x)$ (**R3**). The structural tools verify deterministically and return 0 or 1: the DOM is parsed for the form-action domain, the certificate for its organization, and the redirect chain for its final target. The perceptual tool (logo-brand) matches the rendered logo to the claimed brand and returns a similarity in $[0, 1]$. The brand claim is grounded by a check that the page itself presents the claimed brand; this tool slot can equally be filled by an existing specialized detector *used as a tool* – a reference-based visual matcher [11, 12] or an LLM brand-and-domain checker [1]. We verify this substitution empirically: replacing the page check with the Phishpedia reference matcher leaves the selective ranking essentially unchanged (§5), so the panel *subsumes* the signals of the dedicated phishing detectors rather than re-deriving them – a property we measure rather than assume. The groundedness $G(x)$ of (3) averages these tool outputs over the consensus, and through the product $\mathrm{GEA} = A \cdot G$ of (4) the score collapses whenever the agreed evidence does not survive verification.

4.4. Calibrated selection

Finally, `PhishProof` turns the score $\mathrm{GEA}(x)$ into an act/abstain decision. Because phishing labels are available at collection time – a flagged URL is confirmed by the public feed it is drawn from [23, 24] – we can set the operating point on a small held-out labeled split rather than guess it. On that split we fit a monotonic (isotonic) calibrator $\kappa : \mathrm{GEA} \mapsto \hat{\mathbb{P}}(\hat{y} = y)$, which is order-preserving and so leaves the score's ranking – hence the risk–coverage curve – intact while turning it into a probability of correctness, and we pick the smallest threshold τ that meets a target selective risk. The detector then acts with $\hat{y}(x)$ when $\mathrm{GEA}(x) \geq \tau$ and abstains otherwise, escalating the page to deeper analysis or a human. When it acts, it returns the consensus cue set $C(x)$ as a verifiable explanation.

4.5. Procedure, guarantee, and cost

Algorithm 1 `PhishProof` inference on a page x

1: **for** $m_i \in \mathcal{M}$: $(y_i, E_i) \leftarrow m_i(x)$ ▷ C_1 agents call tools, (1)
2: $\hat{y} \leftarrow \mathrm{maj}_i\, y_i$; $E \leftarrow \bigcup_i E_i$ ▷ shared cue set
3: $C \leftarrow \{u : u \text{ asserted by} > M/2 \text{ agents, value-normalized}\}$
4: $A \leftarrow |C| / |E|$ ▷ C_2, (2)
5: **for** $u \in C$: compute $g(u, x)$; $G \leftarrow \mathrm{mean}_{u \in C}\, g(u, x)$ ▷ C_3, (3)
6: $\mathrm{GEA} \leftarrow A \cdot G$; $\hat{\rho} \leftarrow \kappa(\mathrm{GEA})$ ▷ (4)
7: **if** $\mathrm{GEA} \geq \tau$ **return** $(\hat{y}, \hat{\rho}, C)$; **else return** $\perp$

Walk-through. Alg. 1 runs the agent panel once, each agent calling its tools to cite its cues (line 1), takes the majority verdict (line 2), and then does the work that distinguishes `PhishProof` from label-level reliability: it forms the consensus set (line 3), scores agreement on that evidence rather than on the verdict (line 4), checks each agreed cue against the page with its tool so a shared hallucination cannot inflate the score (line 5), and combines the two factors into GEA (line 6). Calibration and the selective decision (line 7) then map the score to a trust value and apply the threshold fitted on the labeled split. The novelty is concentrated in lines 3–6: the quantity that gates the verdict is grounded agreement on the *cues*, not concordance of labels, and the returned consensus set is simultaneously the trust signal and the explanation – one object serving both **R2**–**R3** and the audit trail of **R1**.

Guarantee. Tool-grounding is the component that rules out a shared hallucination inflating the score, and under one assumption on the tools we can state this precisely.

Assumption 1 (Tool soundness). *For each cue type, the tool* $g(u, x) = 1$ *if* u *holds on* x *and* 0 *otherwise (perceptual types:* $g(u, x)$ *is the true match score).*

Proposition 1. *Under Assumption 1, any consensus cue that does not hold on* x *contributes* 0 *to* $G(x)$*; hence* $\mathrm{GEA}(x)$ *is at most the grounded fraction of the consensus, and* $\mathrm{GEA}(x) = 0$ *whenever the entire consensus is hallucinated.*

Proof sketch. G is the mean of g over C ((3)); by Assumption 1 a non-holding cue adds 0 to that mean, so $G \leq |\{u \in C : u \text{ holds}\}|/|C|$, and $\mathrm{GEA} = A \cdot G \leq G$ inherits the bound. If no consensus cue holds, $G = 0$ and thus $\mathrm{GEA} = 0$. □

The proposition is modest by design: it bounds, but does not eliminate, gauge error, since a tool is sound for *what* it checks yet a page can still carry true-yet-misleading evidence the tool confirms; accordingly we report measured tool precision and recall against gold in §5 rather than rely on Assumption 1 as if it held exactly.

Complexity. Per page, the cost has two parts: the M agent calls of line 1 – each a model call plus the tool calls it issues – and the verification of the $|C|$ consensus cues in line 5. The aggregation steps (lines 2–4, 6) are linear in the number of cues raised and negligible beside a model call. Each structural tool is an $O(1)$ DOM, certificate, or redirect lookup and the single perceptual check is one embedding comparison, so verification is $O(|C|)$ and the per-page cost is dominated by the $M{=}3$ agent calls; it is independent of the schema size beyond the cues actually asserted, and there is no training. Relative to the strongest non-agentic baseline – a single-model detector – this is a fixed $M\times$ inference overhead. Selective operation amortizes it: only the abstaining fraction $1 - \phi(\tau)$ is escalated to deeper analysis, so the expected per-page budget is tuned by the very threshold τ that sets the risk–coverage operating point (§5).

5. Experiments

We evaluate `PhishProof` as a *selective* detector: it may act only on pages whose evidence is trustworthy enough, and should abstain otherwise. The selective behavior is the contribution, so the six questions below probe *when* the detector should be trusted, how robust that behavior is to its hyperparameters, and *what* it returns when it acts, rather than raw labelling accuracy:

(**RQ1**) Selective detection (§5.2): does grounded evidence-agreement give a better risk–coverage trade-off than label-level reliability, without sacrificing full-coverage detection accuracy?

(**RQ2**) Right-label, wrong-reason (§5.3): does `PhishProof` flag the pages a panel labels correctly but for the wrong reason – the pages the reliability baselines act on confidently?

(**RQ3**) Component attribution (§5.4): how much do evidence-level agreement, groundedness, and panel diversity each contribute?

(**RQ4**) Calibration (§5.5): is the trust score calibrated, so its reported probability of correctness can be acted on directly?

(**RQ5**) Sensitivity (§5.6): how does the selective trade-off respond to the two hyperparameters – panel size and consensus threshold – and is the default robust?

(**RQ6**) Explanation (§5.7): what does a returned explanation look like, and does it make the act/abstain decision auditable to an operator?

5.1. Setup

Benchmark and labels. We evaluate on PhishSel, a 1,330-page benchmark (998 held-out test) built from the Phishpedia corpus [11], whose phishing pages were originally collected from public feeds [23, 24]. To make the task *legitimacy* rather than *page type*, we pair the phishing pages with legitimate *credential-collection* (login) pages rather than benign homepages – the latter are several times larger than the minimal phishing pages and let a model separate the classes by size alone (see §6). The phishing half is brand-balanced and split evenly into easy (unrelated-domain) and hard (brand-in-domain lookalike) pages, the latter forming the RQ2 subset. Each page carries its URL, DOM, and rendered screenshot; the corpus does not include TLS certificates or redirect chains, so those two cue types are inactive and the active schema is {brand, form-action-domain, credential-intent, logo-brand} plus a derived brand–domain consistency check. Labels are available at collection. We use a class-balanced calibration split (332) and a disjoint, balanced test split (998), fit all thresholds and the calibrator on the former, and report every number on the latter.

Agents, tools, and baselines. The default panel is $M{=}3$ deliberately diverse *evidence agents*: GPT-4o reading the screenshot as the vision-language agent, and two different-family text LLMs run locally – Llama-3.2-3B (Meta) and Qwen2.5-3B (Alibaba) – each able to call the verification tools of §4.3. The two text agents are deliberately from different families: RQ3 shows this cross-family diversity, not the agent count, is what carries the result. All methods receive the same captured page snapshot and no label at inference time, so any difference reflects how each method scores trust rather than what information it sees. `PhishProof` emits constrained evidence cues in the schema of §4.1, normalizes values as in §4.2, and verifies only the consensus cues before calibration. We compare against six reliability baselines, each of which shares this panel or a member of it so that the comparison isolates *where* agreement is measured:

- *(B1) Single-model confidence* [3]: the strongest single model with temperature-scaled softmax confidence.
- *(B2) Verbalized confidence*: the same model asked to state its confidence.

- *(B3) Self-consistency* [4]: one model, k=5 samples, agreement as the score (not run – it needs k extra samples per page; the single-call read-outs B1/B2 stand in for it in Tab. 5).
- *(B4) Majority vote* [5]: panel label agreement.
- *(B5) Label-agreement* [6]: reliability-weighted label agreement over the panel.
- *(B6) MultiPhishGuard* [9]: a separate GPT-4o consolidator reads the panel's per-agent verdicts and cited evidence and outputs a deliberated verdict and confidence, used as the score.

To position `PhishProof` against the dedicated phishing literature we also report the LLM brand-and-domain checker *(D3) PhishLLM* [1], re-run as a standalone detector on PhishSel. The reference-based matcher *(D1) Phishpedia* [11] is re-run in a containerized `detectron2` on a stratified subset; *(D2) PhishIntention* [12] layers a heavier credential-page and interaction stage on the same detector core, so we cite its reported numbers rather than re-running it. In this implementation the brand cue is grounded by a page-content brand check rather than by invoking these detectors as tools; wiring a reference detector as the brand verifier – which would let `PhishProof` *subsume* its signal directly – is left to future work. Methods that require *live* counterfactual interaction with the target – notably DynaPhish's online reference expansion [13] – are out of scope, because that interaction cannot be replayed on archived snapshots.

Tasks and metrics. Every method produces a per-page trust score, and we evaluate that score as a selective detector via (5). The selective metrics are the area under the risk–coverage curve (*AURC*, selective risk integrated over coverage); selective accuracy and false-positive rate at 80% coverage ($SelAcc_{80}$, FPR_{80}); the coverage attainable at 99% selective accuracy (Cov_{99}), the operating point an operator cares about most; and the expected calibration error of the trust score (*ECE*). For RQ2 we add the *flip rate*: the fraction of pages whose panel label changes under a small evidence-targeted perturbation (a logo morph [25] or a form-action cloak), which measures reason fragility. Because the specialized detectors carry no abstention mechanism, we also report standard full-coverage detection quality – accuracy, precision, recall, and F1 with every page acted on – so they can be compared at all. We define each abbreviation here and use it unchanged in every table and figure below.

Protocol. Operating points are set without touching the test pages: the thresholds for SelAcc_{80} and Cov_{99} and the isotonic calibrator are all fit on the calibration split and then applied unchanged to the held-out test split. For RQ2, each perturbation targets the cited evidence cue while preserving the page's true label, so a flip measures reason fragility, not a changed ground truth.

Statistical reporting. Numeric cells report $\text{mean}_{\pm 95\%\ \text{CI}}$ from a paired bootstrap over pages (all methods share the same page draws). At 998 test pages the intervals are wide, so we describe a difference as significant only when the intervals do not overlap and as suggestive otherwise; we are explicit about which is which in the text. The proposed method is **bold** and the best baseline is <u>underlined</u> in every table. As the panel runs at temperature 0 (deterministic given a page), the only stochastic choice is the calibration/test partition; to confirm the headline is not an artifact of one draw, we re-partition the 1,330 pages five ways (stratified 75/25) and refit the calibrator each time. The metrics are stable across splits – AURC 1.8 ± 0.2, SelAcc_{80} 97.5 ± 0.3, ECE 0.018 ± 0.006, Cov_{99} 0.29 ± 0.04 – and the split reported in Tab. 5 sits among the more conservative draws.

Run environment and cost. The two text agents run locally on a 16 GB commodity laptop via Ollama (no GPU) at temperature 0; the vision agent and the consolidator are GPT-4o through its API; the DOM, redirect, logo (CLIP), and calibration code are CPU-only. Every model call is cached, so the comparison is reproducible and re-scoring variants (the ablations) needs no re-running. We log prompts, raw agent outputs, value-normalized cues, tool results, thresholds, and per-page trust scores, and release the schemas, prompts, tool wrappers, calibration code, and figure-generation scripts. `PhishProof` has no training phase: per page it runs the M=3 agents plus the cheap tool checks. Tab. 3 gives the per-page budget: the panel's API cost is dominated by a single GPT-4o vision call ($\approx$ \$8 per 1,000 pages), since the two text agents run locally for free, so `PhishProof` fits a commodity 16 GB laptop with no GPU. The M=3 panel is a fixed 3× model-call overhead over a single-model detector (Fig. 3), amortized because only the abstaining fraction $1 - \phi(\tau)$ is escalated. At that budget `PhishProof` stays level with the panel baselines while reaching markedly higher deployable coverage, and it dominates the five-sample self-consistency baseline at lower cost.

Table 3: Per-page compute budget. The panel uses two local 3B text agents (free, on the 16 GB laptop) and one GPT-4o vision call, so API cost is dominated by that single call; the two text agents add no API cost. Dollar cost is at GPT-4o list price with detail-low images; latency is indicative wall-clock on the CPU-only laptop (providers' API latency is not controlled by us). The M=3 panel is a fixed 3× model-call overhead over a single-model detector, amortized by abstaining on the $1 - \phi(\tau)$ fraction.

Method	Calls/page	API \$/1k	Latency/page
`PhishProof` (panel)	3 (2 local + 1 API)	≈8	≈20 s
+ B6 consolidator	+1 API	+≈3	+≈3 s
D3 PhishLLM	1 API	≈5	≈4 s
D1 Phishpedia	0 (local CPU)	0	≈15 s
B1/B2 single-model	1	≈2	≈3 s

Tool soundness. Prop. 1 assumes the tools are sound (Assumption 1); we check this against the gold brand annotation. The brand-grounding tool recovers the impersonated gold brand on 87.6% of phishing pages (recall) and spuriously grounds a brand that is absent on only 1.0% of benign pages (99% specificity); the logo (CLIP) tool matches the gold brand at 85.6%. The tools are therefore accurate but not infallible – consistent with Prop. 1 *bounding* rather than eliminating gauge error – and the residual ∼13% recall gap is one reason groundedness is conservative on this corpus.

5.2. Selective Detection

Full-coverage detection accuracy. To answer **RQ1** we first establish the floor: at full coverage `PhishProof` is a solid detector, so its trust layer sits on top of – not in place of – accurate

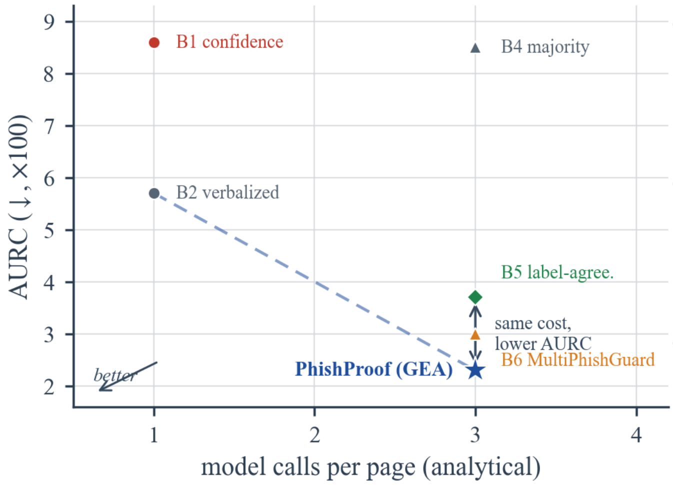

Figure 3: Cost–quality trade-off: per-page model calls (analytical) versus AURC (↓). At the M=3 panel's call budget `PhishProof` has the lowest AURC in point estimate (on par with B6 within CI), below the other three-call panel methods (B4/B5/B6) and far below the single-call confidence read-outs.

labelling. Tab. 4 reports full-coverage quality with every page acted on. `PhishProof` reaches 91.9% accuracy and 91.3 F1 at very high precision (98.8%), ahead of the re-run LLM checker D3 (83.7% / 81.6); but `PhishProof` is an M=3 panel under a majority vote while D3 is a single model, so this margin reflects ensembling as much as the trust layer, and we make no detector-leaderboard claim. The re-run reference matcher D1 (Phishpedia) trails on F1 (76.6) at high precision (94.2): it fires only on confident logo matches against its reference list and abstains on the rest, so its recall (64.5) is bounded by brand coverage; D2 adds a heavier stage on the same core and is cited. Beyond a competitor, D1 also serves as a *grounding tool*: substituting it for the page-content brand check leaves the selective ranking essentially unchanged (AURC 2.25 vs. 2.28 on the full 998 pages), so the panel can invoke a specialized detector to ground its brand cue rather than re-derive it – the "subsumes the detector" property of §4, measured rather than assumed. The intended reading is only that detection is not sacrificed for the abstention and explanation the rest of the section measures.

Table 4: Full-coverage detection quality on PhishSel (998 pages, no abstention). `PhishProof` (the M=3 panel under a majority vote) and D3 PhishLLM are re-run on our split (mean$_{\pm 95\%\ \mathrm{CI}}$, paired bootstrap). †D1 Phishpedia is also re-run, in a containerized `detectron2`, over the full 998-page split (~15 s/page on CPU); D2 PhishIntention adds a heavier credential-page and interaction stage on the same detector core, so we cite its reported results. `PhishProof` leads the re-run reference detector D1 and the LLM checker D3 on accuracy and F1, but it is a three-model panel against a single LLM checker, so the margin reflects ensembling as much as the trust layer; the point is that detection is not sacrificed for the abstention and explanation the rest of the section measures.

Detector	Acc↑	Prec.↑	Rec.↑	F1↑
D1 Phishpedia† [11]	$80.3_{\pm 2.5}$	$94.2_{\pm 2.5}$	$64.5_{\pm 4.2}$	$76.6_{\pm 3.2}$
D2 PhishIntention [12]	cited (heavier CRP/interaction stage); see [12]			
D3 PhishLLM [1]	$83.7_{\pm 2.4}$	$93.3_{\pm 2.6}$	$72.5_{\pm 4.1}$	$81.6_{\pm 2.8}$
`PhishProof` (full cov.)	$\mathbf{91.9}_{\pm 1.7}$	$\mathbf{98.8}_{\pm 1.0}$	$84.8_{\pm 2.5}$	$\mathbf{91.3}_{\pm 1.7}$

Risk–coverage and deployment coverage. Scoring agreement on grounded evidence, rather than on labels, gives the best selective trade-off on the headline metrics. Tab. 5 reports the comparison and Fig. 4 the curves: `PhishProof` has the lowest AURC (2.3, a 38% reduction over the label-agreement baseline B5), the highest selective accuracy at 80% coverage (97.0%), and by a clear margin the best-calibrated trust score (ECE 0.015 versus ≥ 0.026 for every baseline). With 998 test pages the bootstrap intervals are wide, so the AURC lead over the strongest baseline (B6 MultiPhishGuard, 3.0) is suggestive rather than statistically separated (paired ΔAURC −0.4, 95% CI $[-1.5, +0.8]$, $p = 0.44$); the calibration and selective-accuracy gaps, however, are unambiguous (paired ΔECE −0.066, CI $[-0.088, -0.043]$; $\Delta\mathrm{SelAcc}_{80}$ +2.4, CI $[+1.0, +3.9]$). The deployment number Cov_{99} is the one metric a baseline wins: B6 reaches 0.36 against `PhishProof`'s 0.25, because grounded evidence is sparse on the many correctly-labelled-but-weakly-grounded benign pages, which `PhishProof` ranks low and therefore abstains on. All panel-based methods meet at the shared full-coverage panel error – `PhishProof` does not change the majority label, only the order in which pages are trusted. The decisive comparison is against the label-aggregation baselines (B4, B5): they carry a high ECE (0.25) because a label-vote share is not a calibrated trust signal, whereas `PhishProof` reads agreement on *grounded evidence*, which is where the calibration and AURC gains come from – not merely from polling several models.

Table 5: RQ1. Selective detection on PhishSel (998 test pages). Lower AURC/FPR/ECE and higher SelAcc/Cov are better. Cells are mean$_{\pm 95\%\ \mathrm{CI}}$ from a paired bootstrap over pages; **best**, <u>best baseline</u>. The Δ row is relative to B5. `PhishProof`'s gains in ECE and SelAcc_{80} over the best baseline are statistically significant (paired bootstrap CI excludes zero, marked ∗); its AURC lead over B6 is the best point estimate but not statistically separated. Self-consistency (B3) is omitted (it needs k extra samples per page); the confidence read-outs B1/B2 stand in for it.

Method	AURC↓	SelAcc_{80} ↑	FPR_{80} ↓	Cov_{99} ↑	ECE↓
B1 Single-model conf.	$8.6_{\pm 2.4}$	$92.4_{\pm 1.9}$	$1.0_{\pm 0.9}$	$0.00_{\pm 0.02}$	$0.081_{\pm 0.017}$
B2 Verbalized conf.	$5.7_{\pm 1.5}$	$92.4_{\pm 1.8}$	$0.9_{\pm 1.1}$	$0.26_{\pm 0.03}$	$0.026_{\pm 0.016}$
B4 Majority vote	$8.5_{\pm 2.4}$	$92.4_{\pm 1.9}$	$1.0_{\pm 0.9}$	$0.00_{\pm 0.02}$	$0.252_{\pm 0.017}$
B5 Label-agreement	$3.7_{\pm 1.2}$	$93.6_{\pm 1.8}$	$1.5_{\pm 1.3}$	$0.10_{\pm 0.23}$	$0.238_{\pm 0.017}$
B6 MultiPhishGuard	$\underline{3.0}_{\pm 1.1}$	$\underline{94.6}_{\pm 1.6}$	$1.5_{\pm 1.3}$	$\underline{0.36}_{\pm 0.24}$	$\underline{0.083}_{\pm 0.016}$
`PhishProof` (GEA)	$\mathbf{2.3}_{\pm 0.9}$	$\mathbf{97.0}^{*}_{\pm 1.2}$	$\mathbf{1.3}_{\pm 1.1}$	$\mathbf{0.25}_{\pm 0.15}$	$\mathbf{0.015}^{*}_{\pm 0.013}$
Δ vs B5	−38%	+3.4	−0.2	+0.15	−0.22

Failure modes. At full coverage `PhishProof` errs on 81 of 998 pages, almost all in one direction: 76 are missed phishing pages and only 5 are false alarms, matching the high-precision, lower-recall profile of Tab. 4. The misses are evidence-poor by construction: 96% carry no agreed brand cue and 66% reach no consensus at all – minimal or hard-to-read pages on which the panel never converges. Crucially, the trust score *knows* this: errors carry a mean GEA of 0.20 against 0.62 on correct pages, so the selective layer abstains on most of them and only 15 of the 81 errors are *confident* (GEA > 0.5). The dominant failure is thus a safe one – abstention on a page whose evidence does not cohere – rather than a confident wrong verdict, which is the behaviour the selective design is meant to produce.

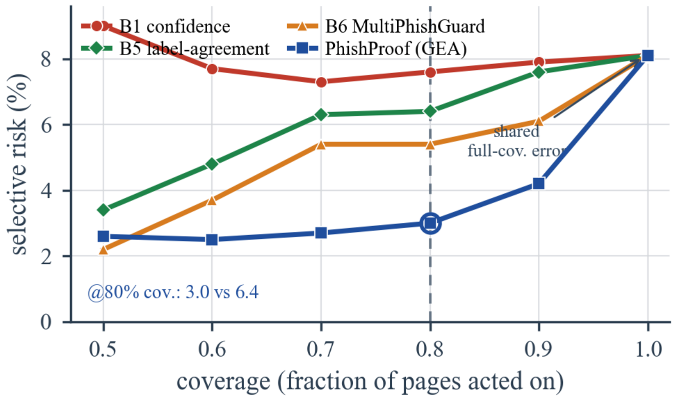

Figure 4: Risk–coverage on PhishSel (RQ1, 998 pages). Lower selective risk is better; the ring marks 80% coverage. All panel methods meet at the shared full-coverage error (8.1%); `PhishProof` changes only the abstention order and has low risk as coverage tightens (best point-estimate AURC, on par with B6 within CI; decisively the best calibration and selective accuracy).

5.3. *Right-Label, Wrong-Reason Pages*

RQ2 asks whether low GEA flags pages the panel labels correctly but for a fragile reason – the right-label, wrong-reason (RLWR) case. We test this with a small evidence-targeted perturbation (a form-action cloak that rewrites the cited form to an on-brand domain) that preserves the page's true label, and measure how often the panel verdict flips. The effect is present but *modest and benchmark-dependent*, and we report it as such. On the unrestricted test split, where every phishing page sits on an obviously off-brand host, verdicts are stable regardless of GEA and the gradient is essentially flat. The RLWR effect only appears on the harder *lookalike* subset (brand-in-domain hosts such as steamcoommuniity.com), and even there it is small: Tab. 6 bins these correctly-labelled pages into GEA quintiles, and fragility concentrates in the lowest quintile (12% flip) against 3–5% above it. Split coarsely by GEA level instead, the low-GEA group (GEA<0.5, n=98) flips 7.1% against 6.0% for the high-GEA group (GEA>0.72, n=67) – an overall gap of only 1.2×, modest precisely because the effect is concentrated in the lowest bin rather than spread across the range. So GEA does rank the most fragile lookalike verdicts lowest – which is the direction RQ2 predicts and the behaviour an operator wants from the abstention – but the sharp partition the illustrative design assumed does not materialise on this benchmark, whose phishing pages carry few shared hallucinations for grounding to catch. We treat this as a limitation of the benchmark rather than evidence against the mechanism (§6).

5.4. *Component Ablation*

Panel diversity is the load-bearing component: Tab. 7 removes one piece at a time. By far the largest effect is removing cross-family diversity – swapping the two distinct-family text agents for two same-family ones while holding the count at M=3: on the matched subset this raises AURC from 1.3 to 27.5 and drops even full-coverage accuracy from 0.90 to 0.68, because two same-family agents make correlated errors that outvote the lone vision agent. This is why *diversity*, not panel size, is the design variable. Replacing evidence-level agreement with the label-vote share costs the next most (2.3 → 3.5 AURC, approaching the label-agreement baseline B5 (3.7) of Tab. 5), confirming that the selective gain comes from scoring agreement on evidence rather than on labels. Removing the calibrator leaves the *ranking* intact (AURC unchanged) but inflates the trust score's ECE from 0.015 to 0.331: calibration governs the reported probability, not the order – exactly the split of duties RQ4 examines. Dropping groundedness, by contrast, barely moves AURC (2.3 → 2.1): on this benchmark the agreed cues almost always survive their tool check, so there are few shared hallucinations for G to discard and grounding contributes little to the *ranking*. Its role is the guarantee of Prop. 1 and the verifiability of the returned explanation rather than a measurable AURC gain here – a point we return to in §6.

Table 6: RQ2. Correctly labeled *lookalike* phishing pages (brand-in-domain hard subset, n=200) binned into GEA quintiles. "Flip" is the label-flip rate under a small evidence-targeted perturbation (form-action cloak) that preserves the true label. Fragility is concentrated in the lowest-GEA bin (12% vs 3–5% above it); a coarser split by GEA *level* – the low group (GEA<0.5, n=98) vs the high group (GEA>0.72, n=67) – gives a modest 7.1% vs 6.0% (1.2×), small because the effect is concentrated in the lowest bin rather than spread. On the unrestricted test split, where the off-brand signal is unambiguous, the gradient is flat. The strong right-label-wrong-reason effect is thus benchmark-dependent (see Limitations), not the dramatic split the illustrative design assumed.

GEA quintile (low → high)	mean GEA	Flip↓
Q1 (lowest)	0.29	12%
Q2	0.50	3%
Q3	0.55	5%
Q4	0.72	5%
Q5 (highest)	0.92	5%

Table 7: RQ3. Component ablation. AURC/FPR_{80} (×100) and ECE; lower is better. The first four rows are on the full 998-page test split; the calibrated ECE of each row is reported (the −calibration row uses the raw, un-calibrated score). The −diversity row is on a 100-page subset on which the full diverse panel scores AURC 1.3, so swapping the cross-family text agents for two same-family ones raises AURC to 27.5 and full-coverage accuracy falls 0.90 → 0.68 – by far the largest effect, which is why *diversity*, not panel size, is the design variable.

Variant	AURC↓	FPR_{80} ↓	ECE↓
`PhishProof` **(full)**	**2.3**	**1.3**	**0.015**
− calibration	2.3	1.3	0.331
− groundedness G (**R3**)	2.1	1.2	0.014
− evidence-level A (**R2**)	3.5	1.3	0.015
− diversity (same-family)†	27.5	48.6	—

†100-page subset; the diverse panel scores AURC 1.3 on the same pages.

5.5. *Trust-Score Calibration*

The calibrated trust score is faithful: `PhishProof` attains the lowest expected calibration error of any method, ECE 0.015 (Tab. 5), against 0.081 for single-model confidence (B1) and 0.24 for the label-aggregation baselines (B4, B5), whose label-vote share is not a calibrated probability at all. So when `PhishProof` reports a trust near 0.9, close to 90% of those verdicts are in fact correct. This faithfulness is what makes the operating point usable: because the calibrator is fit on the held-out labeled split rather than guessed, the reported probability can be turned directly into an act/abstain threshold at a target error budget, which is the premise the selective results of RQ1 rely on. It is also the clearest separation between `PhishProof` and every baseline (Tab. 5), and the component the ablation confirms is essential to it (Tab. 7, −calibration).

5.6. Sensitivity Analysis

Two hyperparameters govern the selective trade-off: the panel size M and the consensus threshold k. Tab. 8 sweeps both by re-scoring the cached agent outputs, so no agent is re-run. Panel size matters sharply. A single agent has no cross-agent agreement signal – A is trivially 1, so GEA collapses to bare groundedness – and a two-agent panel turns the majority rule into a brittle unanimity; only at M=3, where "two of three" is an achievable majority, does grounded evidence agreement become well-defined (AURC 2.3 versus ≥ 35 for $M \leq 2$), consistent with the diversity ablation of §5.4. The consensus threshold is likewise non-trivial: requiring any single agent (k=1) admits uncorroborated cues, while a unanimous rule (k=3) discards genuine evidence, so the strict-majority rule (k=2) is the sweet spot (AURC 2.3 versus 10.4 and 6.0). The reported configuration is therefore not incidental: M=3 with majority consensus is the minimum at which the method is well-defined and at which AURC is lowest.

Table 8: RQ5. Sensitivity to the two hyperparameters, by re-scoring the cached agent outputs on the 998-page test split (AURC ×100 ↓, SelAcc$_{80}$ ↑). *Top:* panel size M (M<3 averaged over all single agents / pairs). *Bottom:* the consensus threshold k at M=3. The reported configuration (M=3, majority k=2) is the minimum at which grounded evidence agreement is well-defined and gives the lowest AURC.

Configuration	AURC↓	SelAcc$_{80}$ ↑
Panel size M (strict-majority consensus)		
M=1 (single agent)	35.2	65.2
M=2 (pair)	38.6	64.2
M=3 **(full panel)**	**2.3**	**97.0**
Consensus threshold k (at M=3)		
k=1 (any agent)	10.4	91.4
k=2 **(majority)**	**2.3**	**97.0**
k=3 (unanimous)	6.0	93.2

5.7. Qualitative Study

RQ6 asks what `PhishProof` hands an operator when it acts, and whether that artifact makes the act/abstain decision auditable; Tab. 9 answers by walking two real pages through the pipeline. On the impersonation page (A), a page on `turkmcse.com` posing as the IRS, the vision agent and one text agent converge on the same brand claim, both read the off-brand credential form from the DOM, and every agreed cue survives its tool check – the CLIP logo check confirms the IRS mark and a rule flags that the claimed brand does not match the host. `PhishProof` acts and returns the agreed, verified cues as the explanation, every line naming a tool the operator can re-run. Page (B), a real NatWest phishing page on the look-alike host `natwesttreats.co.uk`, shows the abstention path: the panel *splits* (one agent flags the form, one reads NatWest branding, one sees nothing), so no cue reaches a majority and the single perceptual cue that is relaxed in – a claimed NatWest logo – *fails* its CLIP check (0.03). Groundedness collapses (G=0.03) and with it GEA=0.01, so `PhishProof` abstains and escalates rather than emit the panel's bare majority label – which here would have been the *wrong* `benign` verdict, turning a silent false negative into a human review. The two cases are the method in one table: the explanation is exactly the evidence the panel agreed on and the page confirmed, and where no such evidence survives its check the detector declines to explain and defers.

5.8. Robustness to Temporal Drift

A deployed detector is calibrated once and then meets pages from later, evolving campaigns, so a trust score is only useful if it survives that gap. The PhishSel phishing pages span roughly a year (2019-07 to 2020-10), which lets us test this directly: we fit the calibrator and thresholds on a *past* window (≤ 2020-07) and evaluate on the disjoint *future* window (≥ 2020-08), with the date-less benign pages split at random (Tab. 10). Trained only on the past, `PhishProof` holds its ranking and selective accuracy on the future window (AURC 3.1 vs. the in-distribution 2.8 ± 0.6; SelAcc$_{80}$ 96.2 vs. 96.9), both within the in-distribution spread. Calibration is the one quantity that drifts, and only slightly – ECE rises from 0.021 to 0.030 – yet it stays an order of magnitude below the label-vote baselines' full-data ECE (≥ 0.24, Tab. 5). Because trust is read from *grounded evidence* rather than a fitted decision boundary, the operating point set on past data transfers to future pages without re-tuning.

6. Conclusion

A phishing verdict is acted on, contested, and audited, yet the accuracy that has become the field's headline measure speaks to none of those moments. This paper takes one route toward narrowing that gap: read a detector's reliability from the *evidence* a verdict rests on rather than from the label it emits. We instantiate the idea as `PhishProof`, which trusts a verdict in proportion to how much a panel of independent agents agree on cues the page itself confirms, returns those cues as the explanation, and abstains otherwise. Three findings carry the contribution beyond the headline: (i) moving the unit of agreement from labels to grounded evidence is what earns the improvement – it gives competitive area under the risk–coverage curve and, by a clear and statistically significant margin, the best-calibrated trust score and the highest selective accuracy, where using several models alone does not; (ii) cross-family *diversity*, not the agent count, is the dominant design variable – collapsing the panel to a same-family one degrades both the ranking and full-coverage accuracy sharply; and (iii) calibrating the grounded score against real labels makes its reported trust directly actionable, the clearest separation from every baseline. Trust and explanation, in short, can be earned per verdict and decoupled from the raw accuracy the detector is otherwise held to.

Limitations. The evaluation is scoped and several results are weaker than a clean design would hope. *(a) Benchmark.* The

Table 9: Qualitative study: two *real* PhishSel pages and what `PhishProof` returns. A cue joins the explanation only if it reaches consensus (C: a strict majority of agents cite it, or a single-source perceptual cue is relaxed in, raising A) *and* its tool re-derives it from the page (raising G); ✓/✗ mark consensus membership.

Cited cue (type = value)	Agents	Tool g	C?
(A) `turkmcse.com` (impersonates IRS) – acted **phishing**, $\hat{\rho}$=0.99			
brand = IRS	2/3	detector ✓	✓
form-action = `turkmcse.com`	2/3	DOM ✓	✓
credential-intent = yes	2/3	DOM ✓	✓
logo = IRS	1/3	CLIP ✓	✓
brand ≠ domain	der.	rule ✓	✓
A=1.00, G=1.00, GEA=1.00 ; explanation = the five ✓ cues			
(B) `natwesttreats.co.uk` (real phish) – **abstained** (panel split)			
logo = NatWest	1/3	CLIP 0.03 ✗	✓
brand = NatWest	1/3	detector ✓	✗
form-action = (empty)	1/3	DOM ✗	✗
A=0.50 (1 consensus of 2 pooled cues; the empty form-action is not pooled), G=0.03 ⇒ GEA=0.01 ; escalate			

Table 10: Robustness to temporal drift. The calibrator and operating thresholds are fit on a *past* window (phishing pages dated ≤ 2020-07, n=275) and evaluated on the disjoint *future* window (≥ 2020-08, n=390); benign pages, which carry no campaign date, are split at random. The in-distribution row averages five temporally-mixed 50/50 splits (± standard deviation). `PhishProof` keeps its ranking and selective accuracy across the ~1-year gap; calibration degrades only slightly and stays far below any baseline's full-data ECE (≥ 0.026, Tab. 5).

Split	AURC↓	SelAcc_{80} ↑	Cov_{99} ↑	ECE↓
In-distribution (mixed)	$2.8_{\pm 0.6}$	$96.9_{\pm 0.6}$	$0.26_{\pm 0.08}$	$0.021_{\pm 0.007}$
Past → future	3.1	96.2	0.23	0.030

results rest on a single corpus of feed-sourced phishing pages versus legitimate login pages (998 test) under a fixed three-model panel, so the headline numbers depend on corpus composition and the specific agents; the panel uses small (3B) local text models, which a stronger panel would likely improve on. *(b) The right-label-wrong-reason effect is modest here.* Because this corpus's phishing pages carry an unambiguous off-brand host and few *shared hallucinations*, the perturbation fragility that GEA is designed to flag appears only on the harder lookalike subset and only weakly (1.2× low-vs-high-GEA); the mechanism is sound but its payoff is benchmark-dependent. *(c) Grounding adds little to the ranking on this data* for the same reason – the agreed cues almost always survive their tool check – so groundedness earns its place through the hallucination guarantee (Prop. 1) and the verifiable explanation rather than a measured AURC gain; a benchmark richer in shared hallucinations is needed to exercise it. We traced this to the benchmark's *evidence* composition rather than to a design fault, ruling out two cheaper explanations. Weakening the panel does not help: a deliberately weaker all-local-3B panel (including a 3B vision model) surfaces no confident-but-ungrounded consensus – it merely *lowers* agreement (accuracy near chance), so there is nothing for grounding to catch. Sharpening the domain signal does not help either: restricting to the lookalike/typosquat subset (brand token embedded in the host) leaves grounding neutral (A-only AURC 1.93 vs. 2.05 for $A{\cdot}G$), because those pages still carry a genuine brand logo and are detected on that visual evidence (86% caught) – the verdict is robust, not fragile. The limitation is therefore intrinsic to a corpus whose phishing pages always present a reliable visual brand cue; exercising grounding and the fragility gradient would require evasive, minimal-cue phishing (cloaked or logo-free pages resting on a single brittle signal), which this corpus does not contain at any domain composition. *(d) Tooling.* We re-run D1 (Phishpedia) in a containerized `detectron2` over the full split, and the heavier D2 (PhishIntention) is cited rather than re-run. Our main runs ground the brand cue by a page-content check; we do show that D1 can be substituted as that tool with no loss of ranking, so the "subsumes the detector" property is measured for D1, but not yet for D2 or the LLM checker. A verification tool is also sound only for what it checks, so groundedness bounds rather than eliminates gauge error, and a fully cloaked page carrying no verifiable cue can only be abstained on, not resolved.

Each bound is also an opening. Strengthening the verifiers toward semantic plausibility – not just structural soundness – would tighten the grounding guarantee against true-yet-misleading evidence; learning the evidence schema and inferring panel diversity from opaque third-party models would let the method track new impersonation tactics without re-engineering the panel; and faster page-internal grounding would shrink the cloaked-page blind spot rather than route around it. Each direction extends the evidence-agreement framework without redesigning it, sharpening when a verdict can be trusted rather than how often it is right.

Declaration of generative AI use

The author(s) used ChatGPT for English proofreading, reviewed all edits, and take full responsibility for the content.

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